

Qatar-1b

R-0643

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230808 · created 2026-07-25T19:35:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460164.7308 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.06 +/- 0.054
Transit depth (Rp/Rs)^2	0.36 +/- 0.65 [%]
Orbital Inclination (inc)	86.64 +/- 2.93
Airmass coefficient 1 (a1)	0.28 +/- 0.025
Airmass coefficient 2 (a2)	1.03 +/- 0.52
Scatter in the residuals of the lightcurve fit is	8.73 %
Best Comparison Star	#2 - [189, 394]
Optimal Aperture	1.81
Optimal Annulus	18.14
Transit Duration (day)	0.069 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.06 $\pm$ 0.054 vs 0.1463 (deviation 1.6 σ)
Duration fit vs published	0.069 d vs 0.0692 d (deviation 0.01 σ)
Mid-time O-C	0.0 min
Depth SNR	1.1
Residual scatter	8.73 %

Light curve

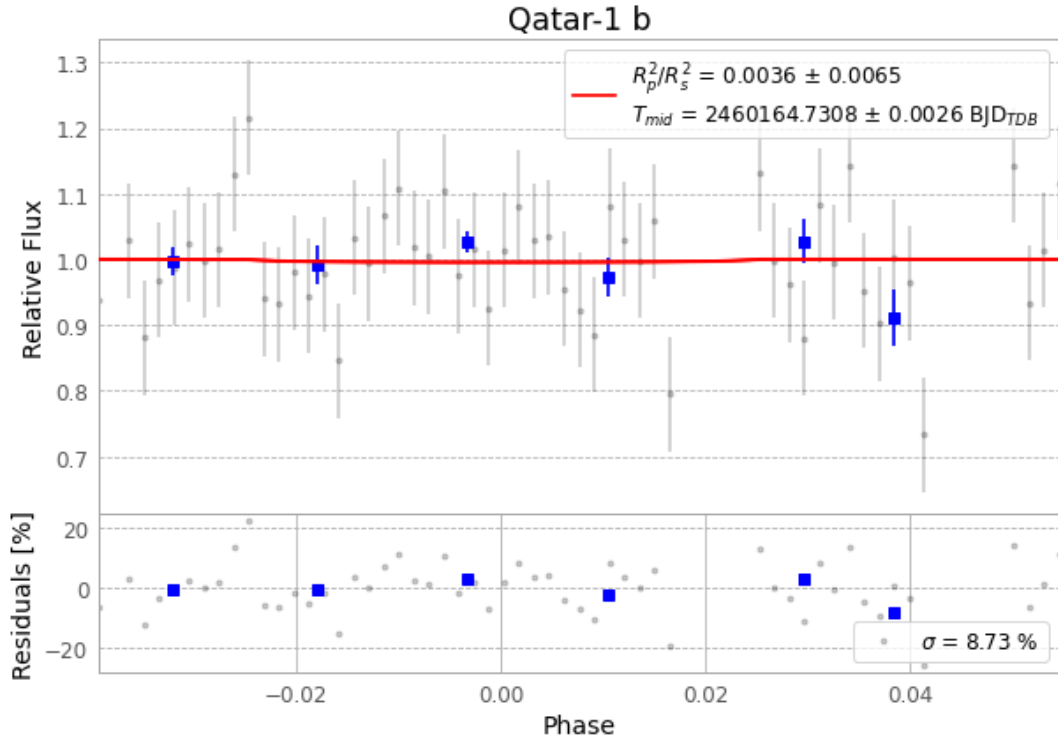


Figure 1 — FinalLightCurve_Qatar-1 b_2023-08-07.png (SHA-256 fde37cae8bba69a2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [230, 222], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 63f9cf6d7dd4885b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

65 FITS frames (43 MB), Qatar-1230808040915.FITS → Qatar-1230808072115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230808.log (4f673bf1d784...), FinalLightCurve_Qatar-1 b_2023-08-07.png (fde37cae8bba...), FinalLightCurve_Qatar-1 b_2023-08-07.csv (4768119c3243...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 19:35Z.